



MOLECULAR BIOLOGY F1 REPORT

**Investigating the role of cellular communication during
cardiomyocyte maturation in the murine heart.**

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Abstract

Background & Aims: Cardiomyocytes (CMs) play a central role in postnatal heart development, undergoing extensive structural, metabolic, and functional changes. These changes are required to support the transition from fetal to postnatal circulation, enabling the heart to meet the increased metabolic and contractile demands after birth. However, CMs lose their regenerative capacity within 7 days after birth. This limited proliferation potential poses a challenge in adult mammalian hearts under disease conditions, often resulting in cardiomyocyte loss and impaired cardiac function. Therefore, understanding the CM maturation as well as their intercellular communication throughout this process is necessary to tackle heart diseases.

Methods: This study investigates the biological process change of CMs during their maturation from day 2 to day 11 of neonatal life and explores the intercellular communication among CMs and other cardiac cell types. We analyzed published single-cell RNA-sequencing (scRNA-seq) of non-CMs and single-nuclear RNA-sequencing (snRNA-seq) of CMs data ranging from the early postnatal window to early maturation stages. Using Seurat, unsupervised clustering was performed to identify distinct cell clusters, which were subsequently visualized with UMAP. clusterProfiler and ReactomePA were used to perform gene ontology and pathway enrichment analyses, providing biological processes underlying cardiomyocyte maturation. CellChat was utilized to examine the intercellular crosstalk between CMs and other cell types during their development process.

Results: Cell populations in neonatal heart tissue were categorized into six distinct categories: two cardiomyocyte clusters, three endothelial cell clusters, two fibroblast clusters, a smooth muscle cell cluster, three immune cell clusters, and an epicardial cell cluster. Moreover, seven clusters of cardiomyocytes were identified and grouped into two maturity stages: an “early stage”- D2D4 and “medium stage”- D9D11. Each stage was associated with different biological processes. Besides, Cellchat analysis revealed time-dependent changes in intercellular signaling mediated by specific ligand-receptor pairs, down-regulated signals from CMs to macrophages through the **ApoE** and **EGF** pathways, decreased signaling between CMs and endothelial cells via **PTPRM**, **NPR1**, and **Netrin** pathways, and reduced CMs–fibroblast communication via **NCAM** signaling. In contrast, **VEGF** signaling between CMs and endothelial cells was upregulated over the same time course.

Conclusions: This study comprehensively characterized the cellular landscape in neonatal heart tissue and revealed the intricate intercellular communication network underlying CM maturation. The stage-specific shifts in CMs’ crosstalk and their associated signaling highlight critical interactions that may contribute to normal cardiac development.

Keywords: cardiac cells, cardiomyocytes maturation, non-cardiomyocytes, scRNA-seq, snRNA-seq.

1. Introduction

According to the WHO, cardiovascular diseases (CVDs) are the leading cause of mortality worldwide, responsible for 19.8 million deaths in 2022, and accounting for 32% of all global deaths [3]. This highlights the urgent need to carry out research into cardiac diseases, especially cardiac fibrosis, to reduce this burden [4]. During the heart failure (HF) progression, cardiomyocytes (CMs) undergo substantial molecular and cellular changes, known as cardiomyocyte remodeling, especially in the ventricles [5]. The remodeling occurs in both physiological and pathological conditions [6]. Physiological remodeling primarily involves reversible CMs hypertrophy to enhance cardiac function [6]. In contrast, pathological remodeling is associated with fibrosis, marked by hypertrophy of surviving CMs, mitochondrial impairment, metabolic irregularities, defective autophagy, followed by irreversible loss of CMs since mature CMs lack the capacity to re-enter the cell cycle for replication [6, 7]. The loss of CMs, mediated in many molecular pathways, such as neurohormonal activation, inflammation response, apoptosis, and necroptosis, is a key event contributing to cardiac dysfunction progression [7]. Therefore, a comprehensive understanding of cardiac cells and their gene expression profiles in the healthy heart during its development and homeostasis can provide valuable insights for disease mechanisms as well as improving therapeutic strategies.

The heart is a hollow, muscular organ and the central component of the cardiovascular system, responsible for pumping blood through the blood vessels [8]. It is divided into four chambers: the right atrium (RA), right ventricle (RV), left atrium (LA), and left ventricle (LV), functioning as two connected pumps that delivery of oxygen and nutrients to the body tissues and the removal of metabolic waste products (**Figure 1A**) [8]. Three layers of the heart wall's tissue, including epicardium, myocardium, and endocardium, harbor heterogeneous cell types that cooperate to ensure proper cardiac function (**Figure 1A**) [9]. Firstly, the outermost layer's epicardium primarily contains epicardial cells (Epis) [9]. The epicardium has been acknowledged as a vital source of multipotent progenitor cells that undergo the epithelial-to-mesenchymal transition (EMT), giving rise to epicardium-derived cells (EPDCs) that differentiate into a variety of cardiac cell types, such as pericytes, fibroblasts, coronary vascular smooth muscle cells, and secrete paracrine signals for cardiogenesis and heart regeneration [10, 11]. Despite being quiescent after birth, the epicardium is reactivated in response to cardiac injury due to upregulated developmental gene programs that

stimulate EMT, thereby assisting the regenerative process [10, 11]. Next, the myocardium or heart muscle, which contains the thickest middle layer, is mainly responsible for contractile function. It is comprised of several cell types, including cardiomyocytes (CMs), smooth muscle cells (SMCs), fibroblasts (Fbs), pericytes (PCs), and vascular endothelial cells (Vas_EC)s [9]. Lastly, the endocardium, the innermost layer that lines the inner surface of the heart chamber, including the heart valve, is made of endocardial endothelial cells (Endo_EC)s [9]. In addition, the heart comprises other cell types, including immune cells, valve cells, and conduction cells. The coordination among these cells is involved in cell-cell interactions (CCIs), paracrine/autocrine signaling, all of which are crucial for maintaining cardiac functions and homeostasis [12].

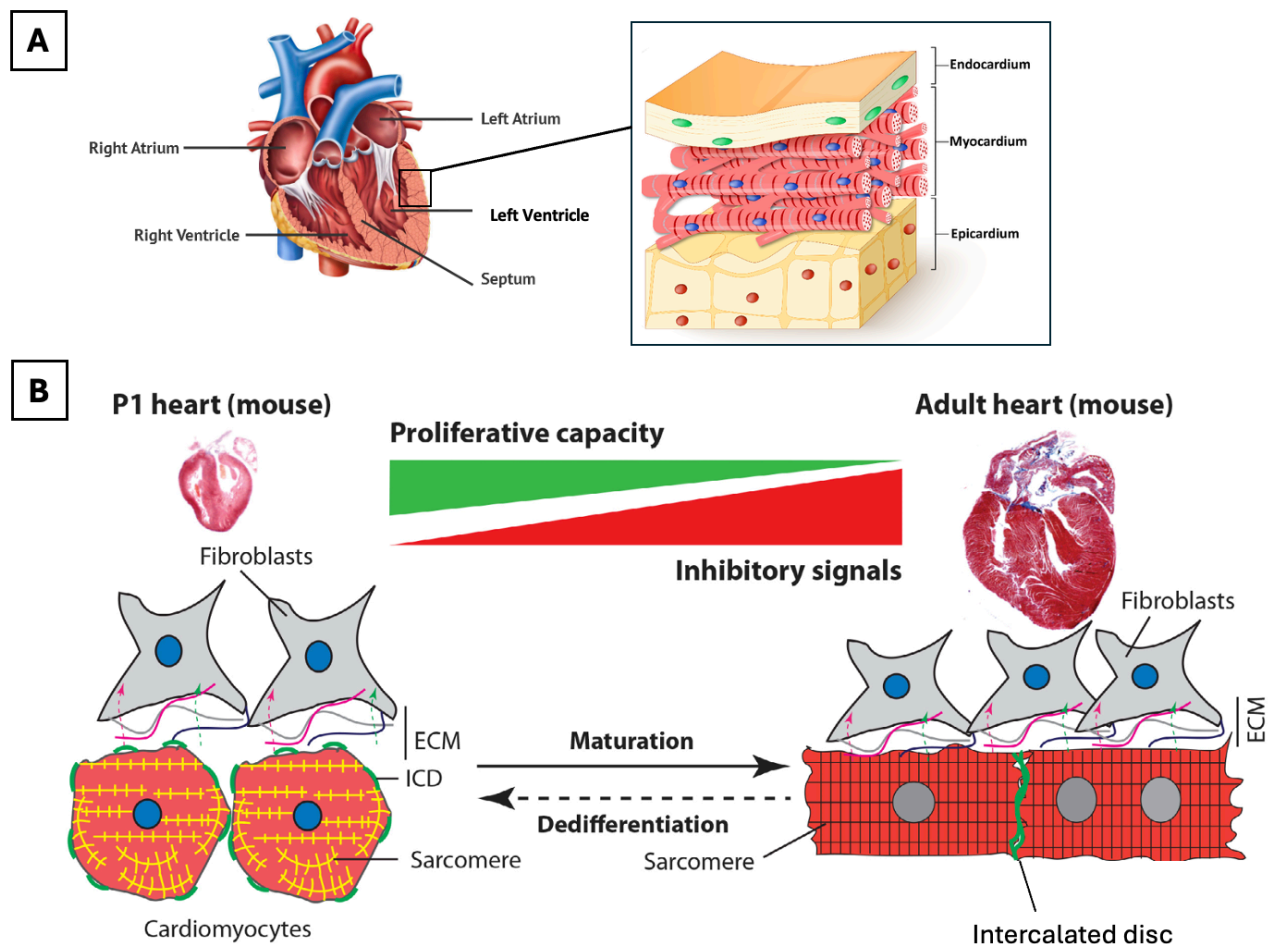


Figure 1: A trans sagittal cross-section of the heart and its wall layers(A) [1], and transitional characteristics of cardiomyocytes from an immature to a mature phenotype (B) [2].

Cardiomyocytes, or cardiac muscle cells, are a specific type of myocyte. They are the primary contractile cells of the heart and contain a high mitochondrial density that generates adenosine triphosphate (ATP) via oxidative phosphorylation to meet the cells' substantial metabolic demands, thereby enabling the heart to pump blood effectively [8]. Sarcomeres, the fundamental contractile units of CMs, consist of interlacing thick and thin filaments of myosin and actin, tropomyosin, titin, and the troponin complex [13]. These sarcomeres are arranged in the repeating structural units that form myofibrils, which are specialized rod-like organelles within CMs (**Figure 1B**) [8]. Although accounting for 20 – 40 % of the cell population, CMs enlarge about 80-90% volume of the heart [14]. Mature CMs are approximately 10–20 μm wide and 50–100 μm long with a striated pattern, an elongated, branched shape, and under involuntary control [8, 14]. The CMs plasma membrane, known as sarcolemma, forms invaginations called transverse (T)- tubules. These T-tubules contain voltage-gated calcium channels that carry action potentials and trigger calcium release from the sarcoplasmic reticulum to initiate excitation-contraction coupling [8]. Cardiomyocytes are interconnected through intercalated discs (ICDs), which consist of gap junctions, adherens junctions, and desmosomes (**Figure 1B**) [14].

Cardiomyocyte maturation is the final phase of heart development, during which fetal CMs undergo extensive changes in gene expression, structural and functional remodelling, and a metabolic shift to achieve a mature phenotype (**Figure 1B**) [15]. This process has been extensively studied in regenerative medicine for recent decades to discover the molecular mechanisms that drive these complex transitions. In neonatal mice, CMs withdraw from the cell cycle within the first week of postnatal life, continuing one more round of DNA synthesis without cell division, with the expression of cell cycle silencing genes such as CDK1, CCNB1, and AURKB [15-17]. This process results in more than 90 % of rodent CMs being binucleated [18]. Whereas, in humans, only about 25% of adult ventricular CMs become binucleated, but rather undergo polyploid nuclei [15, 18]. During this process, levels of cyclins and cyclin-dependent kinases, genes related to promoting cell division, decline while those of cell cycle inhibitors such as p21 and p57 increase [18, 19]. As a result, mature CMs lose their proliferation capacity, contributing to the heart's limited regenerative potential beyond the neonatal stage [16]. During maturation, CMs grow mainly through hypertrophy, with up to a 30-fold increase in cell volume, and organize their sarcomere by the massive expansion and

alignment of myofibrils[18]. Sarcomeric isoform switching due to transcriptional changes or alternative splicing is observed during myofibril maturation [15]. Myosin heavy chain (MYH), the motor protein that drives muscle contraction, switches from fetal MYH7 to adult MYH6 in mice, and vice versa, MYH6 to MYH7 in humans [15]. Additionally, as CMs mature, they switch from glycolysis to fatty-acid energy metabolism to meet increased energy demands, a transition accompanied by substantial mitochondrial biogenesis, resulted mitochondria occupying in up to 40% of the total cell volume [15]. This metabolic shift is orchestrated transcriptional regulators including PPARGC1A, PPARA, and ESRRA, which promote fatty acid oxidation and oxidative phosphorylation [15]. Apart from morphology alterations, CMs maturation also involves functional adaptations, including electrophysiological remodeling and calcium handling, cooperating in enhancing the adult heart's contractile force and endurance [15]. A critical regulator of these processes is attributed to SRF (serum response factor) [15]. The depletion of SRF correlates with the reversed normal maturation patterns in electrophysiological and calcium-handling genes, observed by the upregulation of immature marker *Hcn4*, and the downregulation of mature functional genes such as *Kcnj2* (an inward-rectifier potassium channel), *Serca2a* (the sarcoplasmic reticulum calcium ATPase), and *Ryr2* (the ryanodine receptor 2), as well as defective isoform switching, reduce the global expression level of lipid metabolism, oxidative respiratory, mitochondria biogenesis [15].

Several studies indicate that cell-cell interactions play key roles in CM maturation. For example, in co-culture experiments, cardiac fibroblasts strongly promote the maturation of CMs by forming connexin 43 (CX43) gap junctions [20], and fibroblast subtypes switching is a critical microenvironmental event regulating CM fate [21]. Besides, co-culturing cardiomyocytes with endothelial cells leads to an increased expression level of genes involved in contractile function, such as α -MHC (an isoform of MYH), and improvement of calcium kinetics and sarcomere alignment [18]. In addition, in engineered heart tissue, the interaction of cardiac macrophages with CMs promotes maturation of sarcomeric contractile machinery, mediated via the upregulation of muscle contraction pathways and structural proteins such as troponin, tropomyosin, and MHC [22]. Therefore, elucidating the regulatory pathways that coordinate CM maturation with neighboring cell types is a major challenge but is essential for developmental biology and regenerative medicine.

Modern techniques, such as single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics, are providing high resolution to map these dynamic cellular changes and interactions.

In this study, we investigate the role of intercellular communication during the cardiomyocyte maturation process in neonatal mouse hearts. The study aims to provide insights into cellular crosstalk and the molecular signaling pathways that govern the final steps of cardiomyocyte differentiation.

2. Methods

2.1. Data collection and data analysis workflow

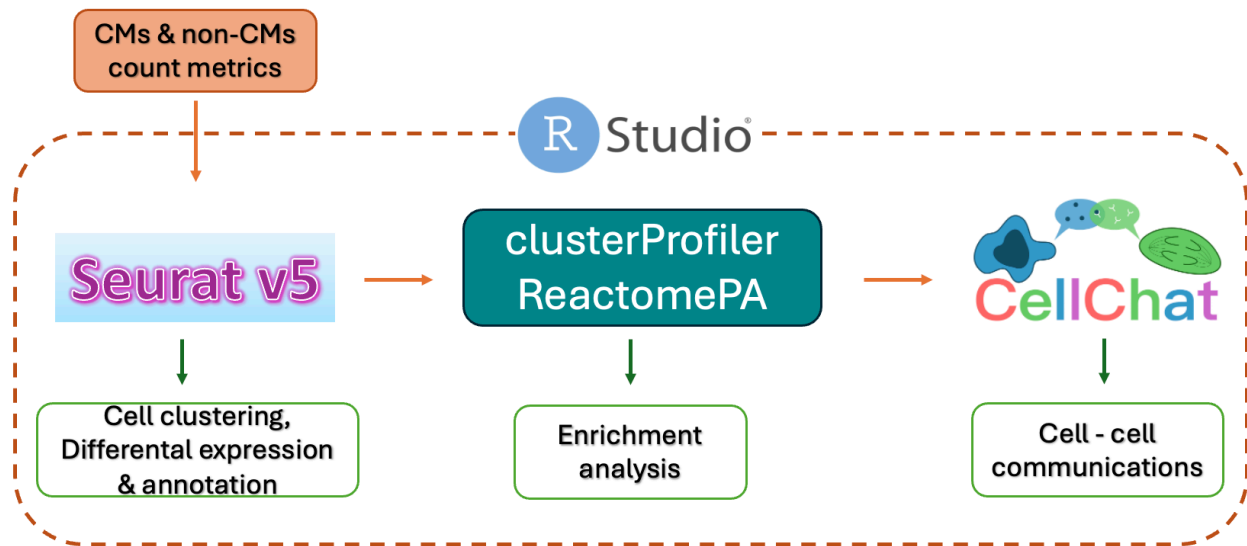


Figure 2: The data analysis pipeline and libraries were used in the current study.

We combined published scRNA-seq and snRNA-seq datasets of CMs and non-CMs from Wang et al., published in 2020 [23]. The data were obtained from the control group, which underwent Sham surgery on postnatal day 1 and day 8. Ventricular cells were collected at days 1 and 3 post-surgery for each postnatal stage and subjected to both scRNA-seq for non-CMs and snRNA-seq for CMs. In total, four neonatal mouse timepoints from the sham group - D2, D4, D9, and D11 after surgery - were included in the analysis.

All analyses were conducted in R, an open-source programming language, using RStudio with the corresponding available packages, including Seurat v5[24], clusterProfiler [25], ReactomePA[26] and CellChat v2 [27] as shown in **Figure 2**.

2.2. *Data preprocessing and principal component analysis*

Seurat v5.2.0 [24] was utilized for processing RNA-seq data, starting by creating the Seurat object from a unique molecular (UMI) count matrix. Quality control was conducted to remove low-quality cells, including those with more than 5% mitochondria and higher than 2500 or fewer than 200 expressed genes, which typically represent cell deaths, doublets, or technical artifacts. Following filtering, gene expression values per cell were normalized using log-normalization, where raw counts were scaled to the same sequencing depth and log-transformed, thereby reducing technical variability while preserving biological signals. To capture the most informative transcriptional features, highly variable genes were identified by modeling the mean–variance relationship across all genes, with the top 2,000 genes selected by default for downstream analysis. Prior to principal component analysis (PCA), a linear transformation, known as ‘scaling’, was performed to shift gene expression to a mean of zero and scale variance to one. This step ensures that highly expressed genes do not dominate downstream analyses by giving equal weight to all genes.

2.3. *Cell clustering and annotation*

Principal components with high standard deviations were selected for further dimensional reduction, as these capture the most biologically significant variation in the data, reduce technical noise, and enhance the reliability of clustering and visualization. Next, cell clustering was performed using a graph-based algorithm, creating clusters that reflect similarities in the transcriptional profiles of individual cells. To visualize cell clusters, the Uniform Manifold Approximation and Projection (UMAP) technique was applied, which learn the underlying structure in the dataset by placing cells with similar transcriptional patterns close together in low-dimensional space. Differentially expressed marker genes for each cluster were identified using the Wilcoxon rank sum test, with thresholds of $\text{avg_log2FC} > 1$ and adjusted p-value ≤ 0.05 . Cell annotation was conducted based on distinctive markers reported in literature reviews.

2.4. *Enrichment analysis of differentially expressed genes*

The aim of an enrichment analysis is to evaluate a collection of related genes, known as gene sets, in order to determine if the genes in these sets are enriched for differential expression. Gene set enrichment analysis (GSEA) was performed with marker genes from cell clusters to identify the

potential biological functions associated with cells within clusters. Differentially expressed genes (DEGs) across timepoints were identified in the Seurat package with a threshold of $\text{avg_log2FC} > 1$ and adjusted p-value ≤ 0.05 . All DEGs were then mapped to Entrez gene identifiers using the mouse annotation package `org.Mm.eg.db`, and subsequently these Entrez IDs were used as inputs for enrichment analyses. Over-representation analysis was carried out by the `clusterProfiler` package via Gene Ontology (GO) Biological Process terms (p-value cutoff = 0.01) [25]. GSEA was also performed for GO terms with a p-value cutoff of 0.05. To further explore biological pathways, pathway enrichment analysis was conducted with `ReactomePA` package [26], with a p-value cutoff of 0.2.

2.5. *Intercellular communication analysis*

The `CellChat` v2 package [29] was employed to predict and visualize statistically significant cell-cell communication among cell clusters representing early and medium stages of CMs development by capturing intercellular communication patterns across cell groups by quantitatively characterizing signaling networks using graph-theory-based metrics. A Seurat object was used as input to generate a `CellChat` object. `CellChatDB`, which incorporates 2,021 validated ligand–receptor interactions, including multimeric complexes and cofactors such as agonists, antagonists, and co-stimulatory molecules, was applied to calculate communication probabilities and identify significant interactions ($p < 0.05$). Pathway-specific intercellular communications were then inferred from the predicted communication probabilities. The number of cell-cell interactions and overall communication strength were visualized using circle plots and chord diagrams. Lastly, the resulting `CellChat` objects for cells in “early stage” and “medium stage” of CMs development were saved individually and subsequently integrated as an input for multiple dataset analysis, enabling direct comparison of signaling patterns between CMs developmental stages.

3. Results

3.1. Cellular heterogeneity in the neonatal heart tissue

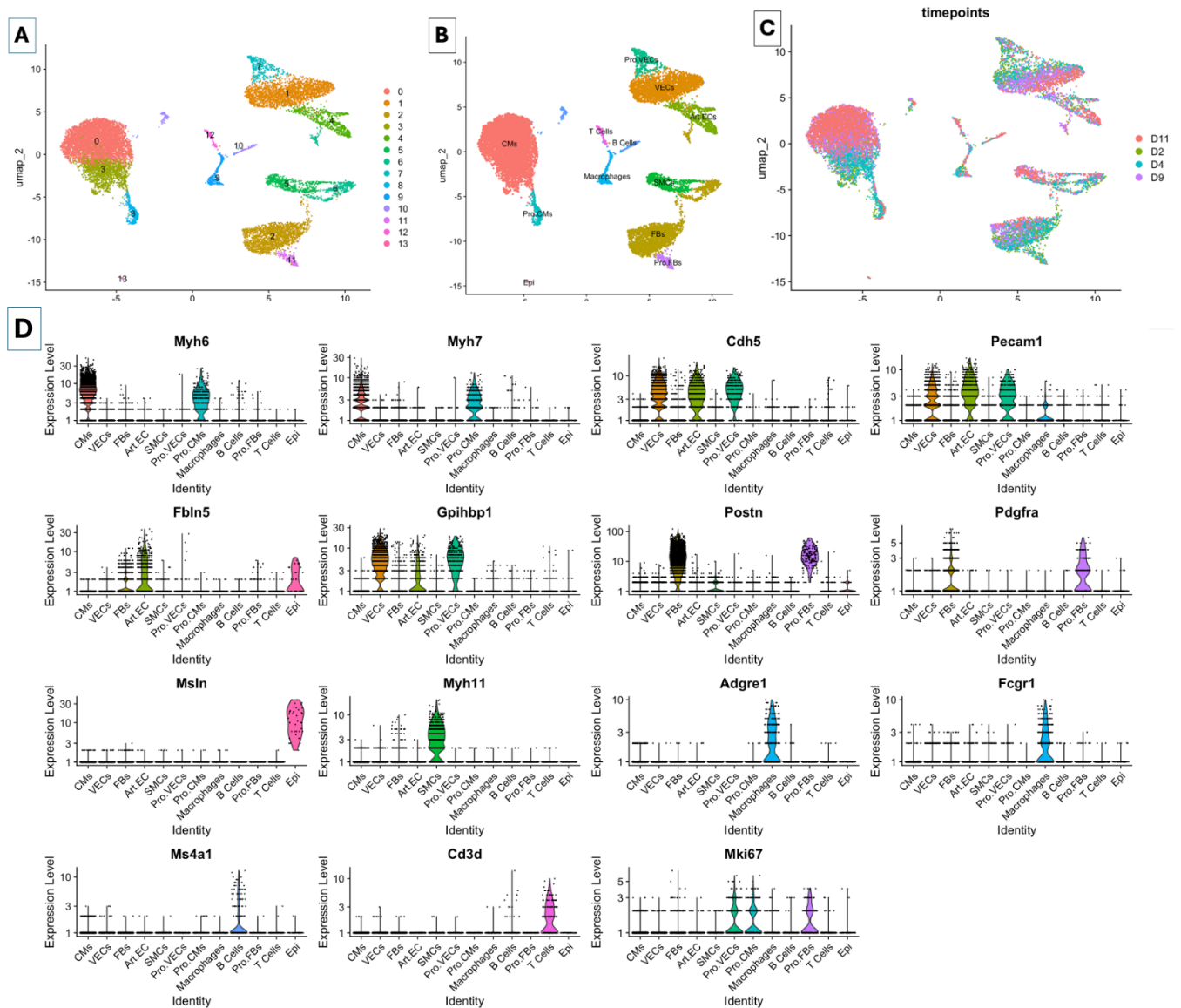


Figure 3: Distinct cellular compositions in neonatal heart tissue. **(A)** Cell clustering reveals the separation of major cell populations. **(B-C)** Cell annotation of identified clusters. **(D)** Violin plots depict the expression of canonical marker genes across clusters.

After quality control, a total of 13,863 cells from all timepoints were clustered into 14 clusters and subsequently annotated into 12 major cell types based on the expression of canonical marker genes (**Figure 3A,D**). Cells that cluster in proximity indicate shared functional roles and transcriptional profiles. These included cardiomyocytes (CMs; *Myh6*, *Myh7*), proliferating CMs (pro.CMs; *Myh6*, *Myh7*, *Mki67*), three endothelial subclusters- (ECs), arterial endothelial cell (Art.ECs; *Cdh5*, *Pecam1*,

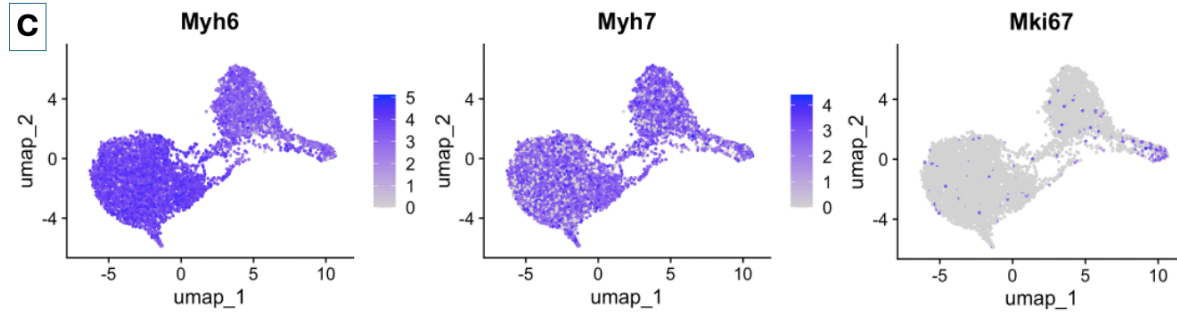
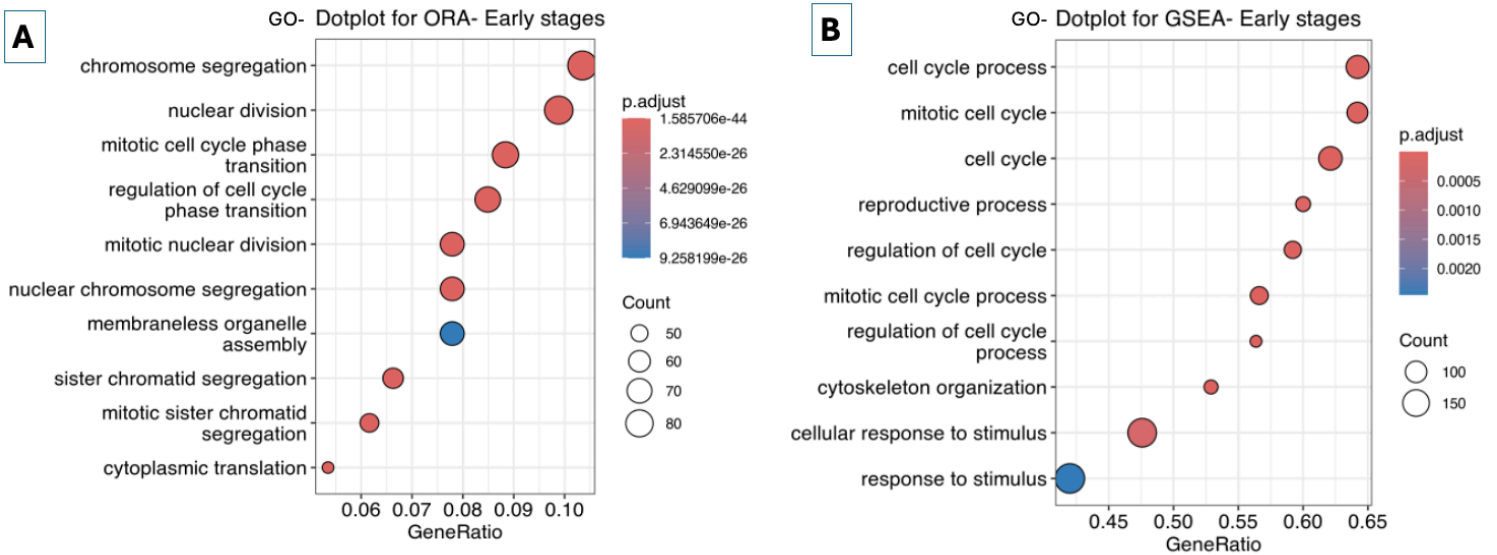


Figure 4: Subclusters of cardiomyocytes (CMs). **(A)** Distinct CM subpopulations clustering results. **(B)** Distribution of CM clusters across different timepoints. **(C)** Expression marker genes specific to CM subclusters. **(D)** Heatmap of differentially expressed genes (DEGs) across CM subclusters. **(E)** Stage-specific DEGs during CM maturation.

To investigate the change of CMs during maturation, CMs were subclustered, resulting in 6,460 cells being obtained after quality control steps. Seven clusters of CMs (CM1-CM7) were identified (**Figure 4A**). The temporal distribution of the CM subclusters was visualized in **Figure 4B**, which displays their dynamic composition across developmental timepoints. To explore transcriptional changes underlying these transitions, differentially expressed genes (DEGs) were identified for each CM subcluster shown in the heatmap (**Figure 4D**). Subsequently, DEGs distinguishing all maturation stages of CM's subpopulations were examined to elucidate the shift in functional roles of CMs in early vs. medium stages. The top 10 highly expressed genes for each stage were visualized in **Figure 4E**. Notably, group D2 and D4, corresponding to CM2, CM4, and CM6, were distributed closely to each other and were grouped as D2D4 timepoint. In addition, clusters CM2, CM4, and CM6 shared the strong expression of genes associated with immature CMs states, such as *Tnni1*- a fetal isoform precursor during isoform switching to mature CMs in ventricular [15], *Ptprg*- involved in cardiac morphogenesis [28], and *Cox7a2*- related to cardiac mitochondrial respiration [29], indicating early stage development of CMs , belong in to D2D4 group (**Figure 4D**). A similar pattern of the proximity in distribution was observed in group D9 and D11, corresponding to CM1, CM3, CM5, and CM7, which were grouped as D9D11 timepoint. Notably, CM4, a subgroup of the early stage, uniquely upregulated mitotic and cell-cycle genes (*Top2a*, *Cdca2*, *Kif11*, *Esco2*), consistent with a proliferative phenotype [30]. High expression of *Mki67* (**Figure 4C**) was correlated with the dense distribution of proliferative CMs in the CM4 cluster. Similarly, early-stage CMs broadly express cell-cycle and DNA-maintenance genes such as *Ttk*, *Esco2*, *Ankle1* [31-33]. By contrast, CM7, a subgroup of the medium

stage, highly expressed transcription factors, including , Fli1, Runx3, Ebf1 and membrane regulators , such as *Gimap6*, highlighting a composite program of immune/autophagy signaling [34], endothelial/vascular regulation [35, 36], and transcriptional control [37]. Additionally, medium-stage CMs upregulated for genes related to ion-handling, electrophysiological functions, including *Trpc4*- involved in Ca^{2+} influx and neonatal cardiomyocyte remodeling under $\text{G}\alpha\text{q}$ protein-coupled receptor (GPCR) stimulation [38], *Hcn2*- genes encoding the pacemaker channel contributing to cardiac electrophysiology [39], *ATP2a3*- gene encoding one of the calcium pump SERCA [40], consistent with electrophysiology and calcium-handling properties of mature CM [15].

3.3. Enrichment analysis of CMs DEGs



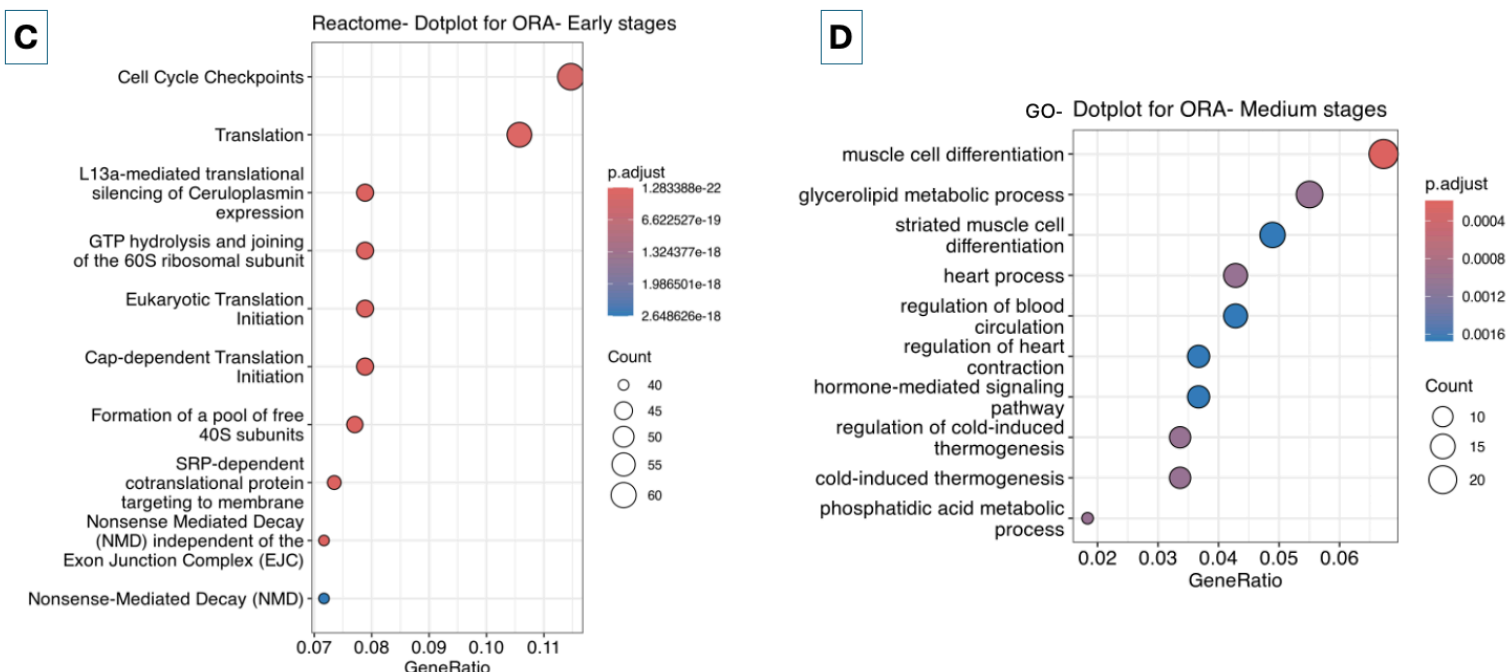


Figure 5: Representative GO (A,B,D) & REACTOME (C) enrichment analysis results of CMs development in early vs. medium stage.

GO Biological Process enrichment analysis revealed that upregulated genes of CMs in the early stages were mostly associated with the cell cycle regulations, mitosis, nuclear division, sister chromatid segregation, and cytoskeletal organization (**Figure 5A, B**). In addition, Reactome pathway analysis further confirmed enrichment in cell cycle checkpoints and translation machinery during early developmental stages, suggesting active protein synthesis to support rapid proliferation and essential processes such as sarcomere assembly [15] (**Figure 5C**). In contrast, gene sets upregulated in medium-stage CMs were related to muscle cell differentiation, heart processes, metabolic adaptation, and thermogenesis, indicating the transition from a proliferative state to functional and structural maturation (**Figure 5D**). However, no pathway reached statistical significance in the medium-stage group under p-value cutoff 0.2.

3.4. Cellular communication during CMs maturation.

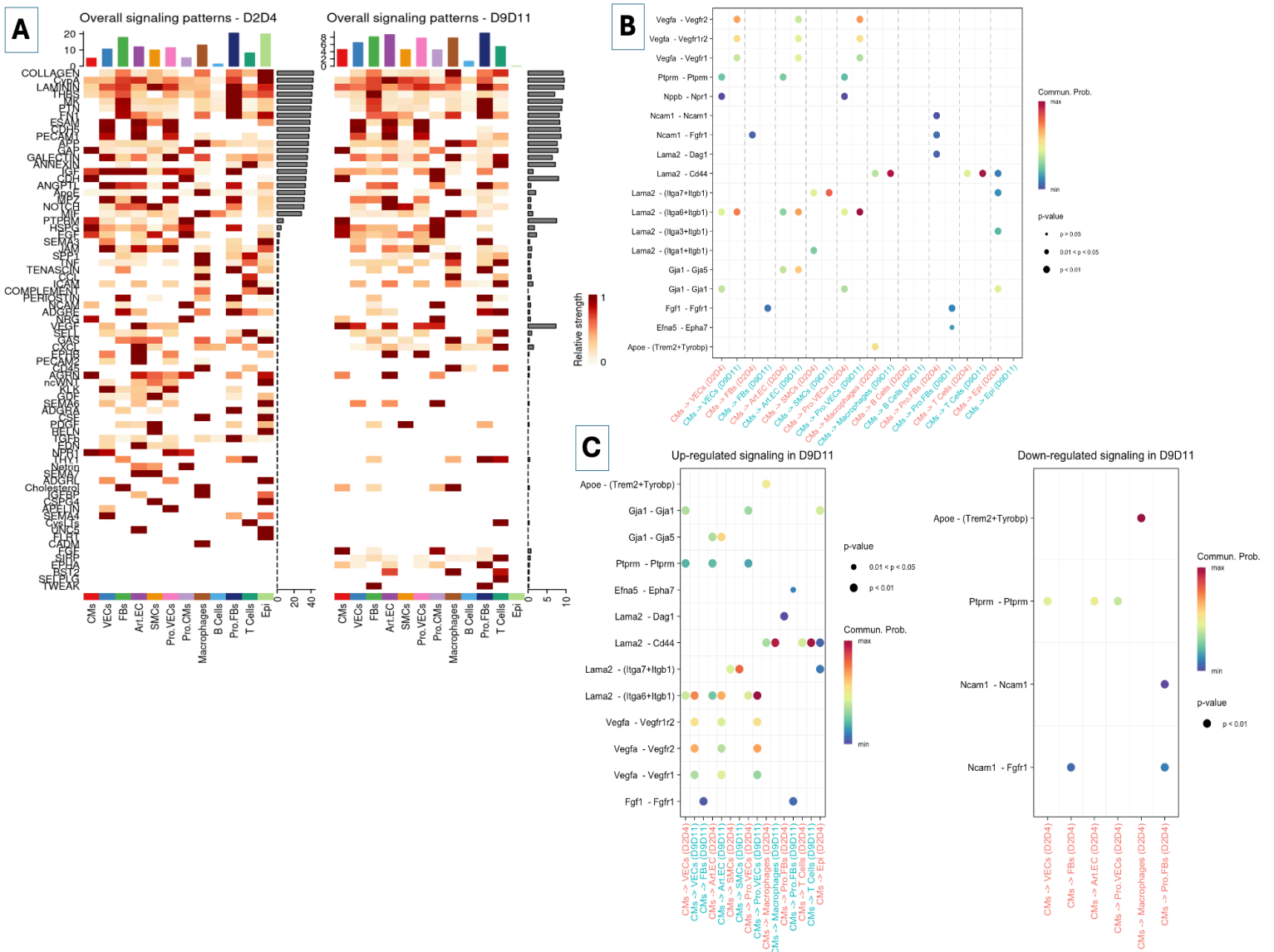


Figure 6: CellChat analysis of a difference in signal communication in neonatal heart tissue between D2D4 and D9D11. **(A)** Heatmap showing overall signaling strength between cell populations in D2D4 and D9D11 groups. The color bar plot on top displays total signaling strength of specific cell clusters in all pathways. **(B)** Scatter plot of communication probability of significant ligand-receptor pairs between CMs and other cell clusters in D2D4 and D9D11 groups. Dot color shows communication probability and dot size depicts computed p-values. **(C)** Bubble plot comparing up-regulated and down-regulated significant ligand-receptor pairs between CMs and other cell clusters in D9D11 vs. D2D4.

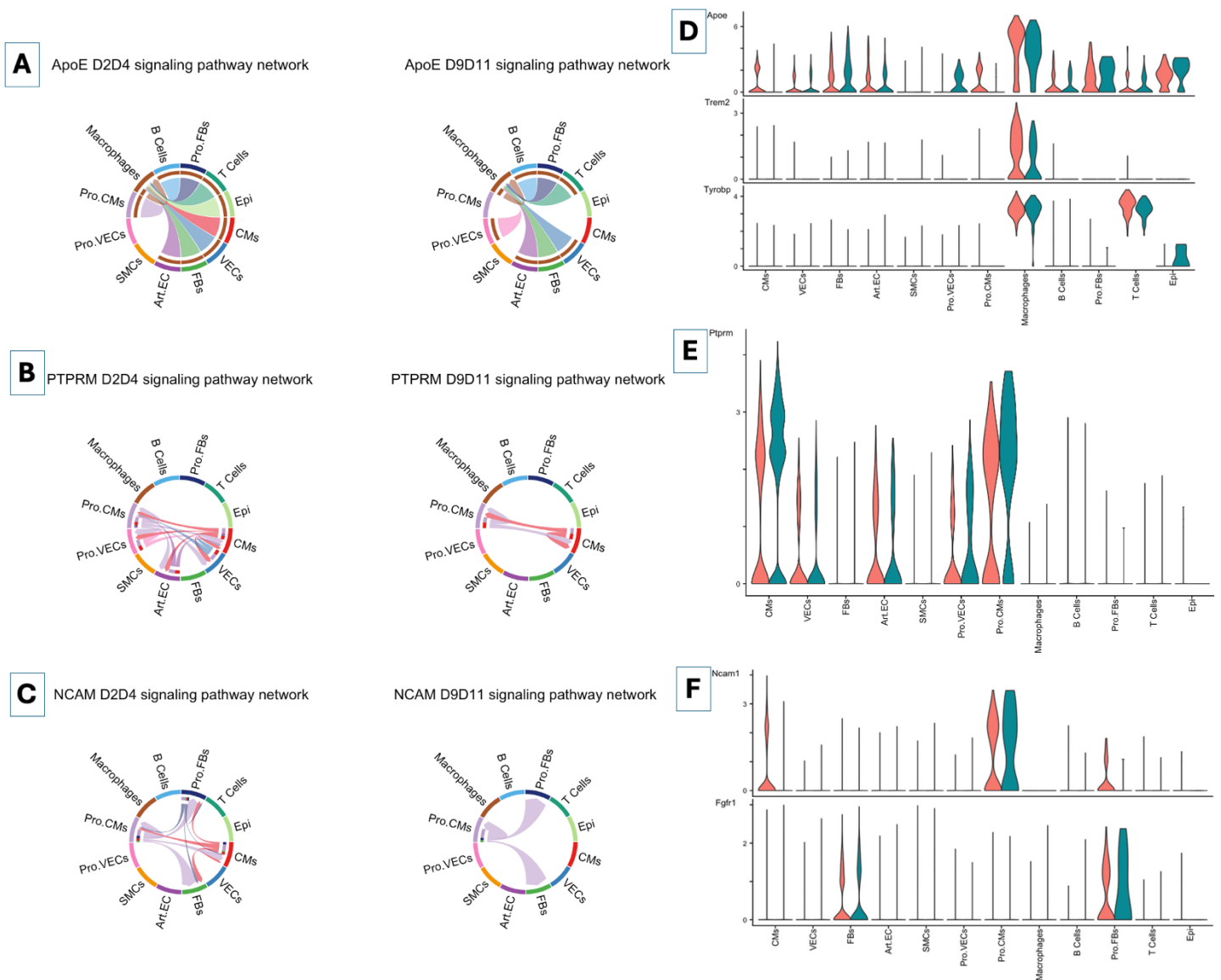


Figure 7: Pathways impact CMs' maturation. (A-C) Circle plots of the interaction on **A)** ApoE, **B)** PTPRM, **C)** NCAM pathways among CMs and other cell clusters, where thickness of the line represents greater communication probability, and color of the line denotes origin of the signal. **(D-F)** Violin plots of the distribution of gene expression of ligand-receptor pairs of the chosen pathway, where orange and green depict group D2D4 and D9D11, respectively.

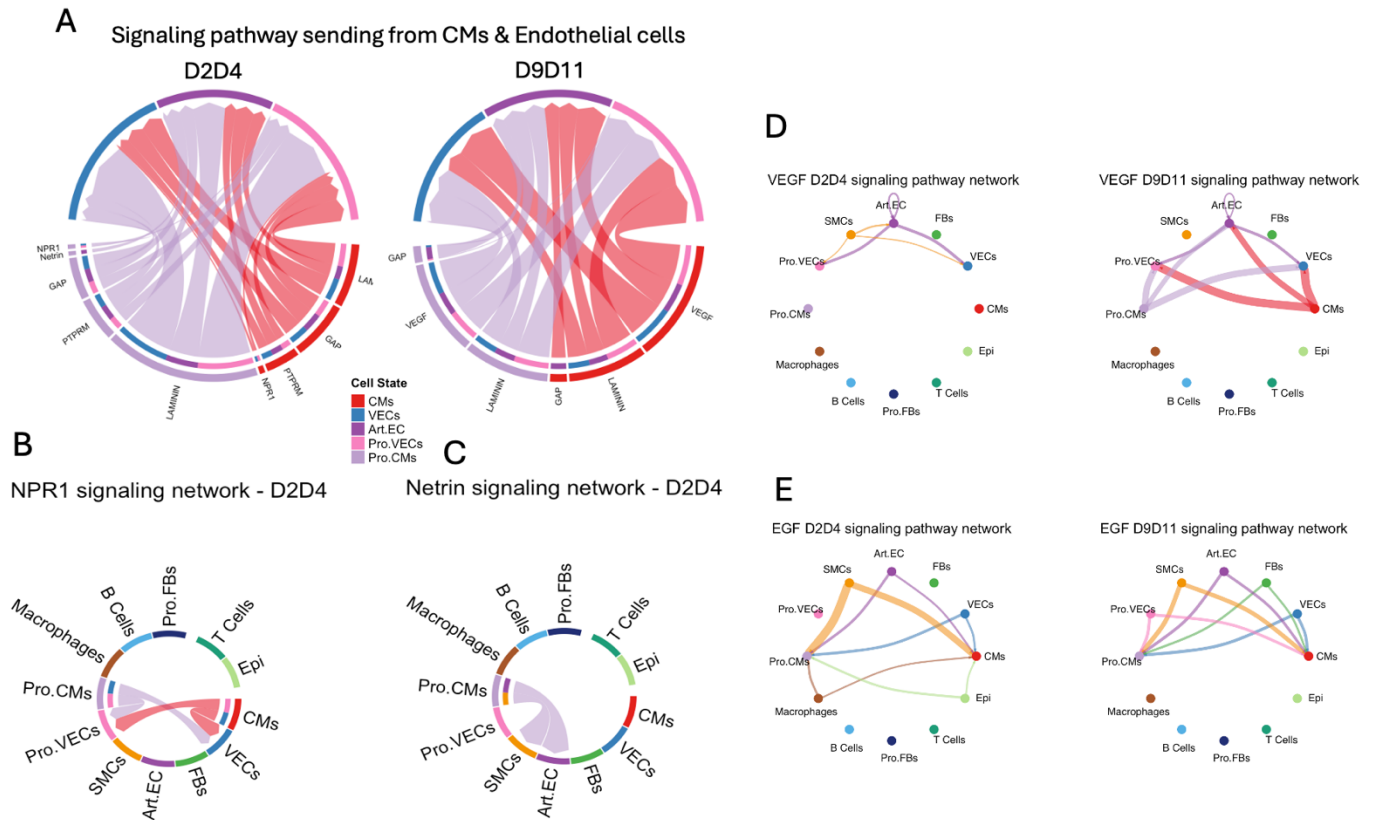


Figure 8: Other pathways contributed to CM's maturation. **(A-C)** Chord diagram of intercellular communication between CMs and endothelial cells. Individual pathway **(B)** NPR1 and **(C)** Netrin that were missed in D9D11. **(D-E)** Circle plots of signaling changes between D2D4 and D9D11 groups related to CMs and other cell clusters.

Ligand–receptor (L–R) interaction analysis using CellChat was performed to compare the signaling network of the intercellular environment during CM's development between the early and medium stages. Communication probabilities were first summarized to evaluate overall differences in information flow **(Figure 6A)**. At the early stage, the top bar plots displayed the maximal interaction near 20 arbitrary units, suggesting widespread intercellular signaling. As development proceeds to D9D11, the diversity and magnitude of signaling events declined, with peak interactions dropping below 10. Notably, extracellular matrix- related pathways such as COLLAGEN and LAMININ remained active, but exhibited reduced signaling strength, while vascular signaling pathways like VEGF became upregulated. These patterns suggest a development transition from broad intercellular communication in early stage towards more specialized and functional signaling profiles in later

stages. Moreover, several signaling pathways originating from or targeting epicardial cells were substantially downregulated at D9D11. In cardiomyocytes, signaling activity also shifted, with CypA, IGF, EGF, NCAM, NRG, and NPR1 pathways being downregulated, while VEGF and FGF signaling were upregulated.

To further explore intercellular communication, the probabilities of L-R pairs mediating CM-to-non-CM interactions were compared between D9D11 and D2D4 (**Figure 6B**). Stronger interactions in D9D11 between CMs and pro.VECs (*Lama2-ITGA7/ITGB1*), CMs and macrophages (*Lama2-Cd44*), and CMs and T cells (*Lama2-Cd44*) was displayed in **Figure 6B**. Differential expression analysis (DEA) of L-R pairs between D2D4 and D9D11 identified several stage-specific interactions (**Figure 6C**). For instance, the ApoE pathway between CMs and macrophages (via *ApoE-Trem2/Tyrobp*), the PTPRM pathway between CMs and pro.VECs (via *Ptpm-Ptpm*), and the NCAM pathway between CMs and fibroblasts (via *Ncam1-Fgfr1*) were strong at D2D4 but diminished over time. The cell-cell communication of these pathways and their corresponding L-R gene expression patterns were further visualized in **Figure 7**.

In addition, signaling pathways involving CMs and endothelial cells were found to disappear during this period (**Figure 8A**). Specifically, NPR1 signaling mediated by the NPPA/BNP-NPR1 L-R pair between CMs and VECs, as well as NTN1-UNC5B/CD146 - mediated Netrin signaling between Pro. CMs and Art. EC, was lost at D9D11, indicating decreased intercellular communication with increased maturation (**Figure 8B, C**) [41, 42]. Moreover, VEGF signaling is fundamental to the formation and maintenance of blood vessels [43], became upregulated at later timepoints, particularly in signals from CMs to Art.EC, VECs, which were absent during the early stages (**Figure 8D**). EGF signaling was detected between macrophages and CMs as well as between epicardial cells and pro.CMs at D2D4 but were also lost by D9D11 (**Figure 8E**).

Together, the dynamic changes of the ligand-receptor interactions between CMs and macrophages, fibroblast, and endothelial cells by D9D11 suggests a stage-specific of intercellular communication networks.

4. Discussion

Cardiomyocyte maturation is fundamental to heart development and disease due to their cell-cycle exit around 7 days after birth. While extensive research has investigated the molecular mechanisms underlying this process, the role of the microenvironment and intercellular crosstalk remains less well defined. In this study, we analyzed neonatal heart transcriptome data to characterize cell–cell communication during CM development. We identified CM4, a cardiomyocyte subpopulation with high expression of *Mki67* and other cell-cycle–related genes, suggesting a proliferative state of CMs, particularly during the regenerative period in neonatal hearts, in line with previous studies [30, 44]. Gene set enrichment analysis further supported this by showing that early-stage CMs were significantly enriched for cell cycle and mitotic pathways. Additionally, other CM subpopulations represented distinct functional and maturation states correlated with cardiac development. Early-stage clusters including CM2 and CM6 also exhibited immature CMs markers, such as the fetal troponin isoform *Tnni1* in myofibril, and genes involved in cardiac morphogenesis- *Ptprg* and oxidative phosphorylation of *Cox7a2*, which similar with other findings [15]. Conversely, medium-stage clusters, exemplified by CM7, displayed upregulation of transcription factors and membrane-associated regulators implicated in immune signaling, autophagy, and vascular interactions, consistent with GO enrichment biological process of muscle cell differentiation, cardiac function, and metabolic processes in medium-stage CMs.

Intercellular communications between CMs and non-CMs, including macrophages, endothelial cells (ECs), and fibroblasts (FBs), are crucial for regulating cardiac homeostasis, remodeling, and response to injury, and may also drive CM maturation[6, 21, 22]. For instance, ApoE–Trem2/Tyrobp signaling between macrophages and cardiomyocytes was enriched at D2D4 (**Figure 7A, D**). This observation is consistent with prior work showing that macrophages secrete CLCF1 to promote neonatal CM proliferation [23]. Notably, ApoE–Trem2/Tyrobp signaling is known to be involved in macrophage-specific activation and anti-inflammatory phenotypes during cardiac injury and metabolic diseases, e.g. myocardial infarction, atherosclerosis, and obesity [45-48], offering that macrophages in early postnatal hearts could support CM proliferation, structural remodeling, or metabolic transition. Supporting this, a study have shown that preserving embryonic-derived macrophages in the adult heart by inhibiting monocyte recruitment reduces inflammation and enhances tissue repair [49].

Moreover, Aurora et al., in 2014, demonstrated that neonatal macrophage depletion leads to fibrotic scar formation, impaired angiogenesis, and reduced cardiac function [50], highlighting the essential role of macrophage-derived signals that promote angiogenesis and regeneration during early heart development [50].

PTPRM, encoding a transmembrane protein tyrosine phosphatase involved in cell growth, differentiation, and mitotic cycle regulation [28], showed downregulated *Ptprm*–*Ptprm* signaling between VECs, arterial ECs, and CMs in this dataset (**Figure 7B, E**). This contrasts with the findings of Wang et al., who reported localized *Ptprm*–*Ptprm* activity in adjacent CMs and capillary ECs from postnatal day 7 onward in the murine heart [51]. This discrepancy could arise from methodological differences, including cell-type annotation or clustering resolution.

Cardiac fibroblasts are essential for maintaining the extracellular matrix and regulating myocardial growth and repair through paracrine signaling and direct CM contact [12]. In our analysis, *Ncam1*–*Fgfr1* signaling between CMs and fibroblasts was decreased by D9D11 (**Figure 7C, F**), suggesting a stage-specific role in early cardiomyocyte development. This aligns with studies showing NCAM-1 as a key adhesion molecule in the cardiac conduction system, highly enriched in cardiac Purkinje cells by postnatal day 7 [52]. Although Purkinje cells represent less than 1% of ventricular cardiomyocytes, they play an indispensable role by enabling synchronized electrical impulse transmission across the ventricular myocardium [52]. NCAM-1 supports ventricular conduction and structural integrity of the His-Purkinje system, with its knockout leading to impaired conduction and morphological abnormalities, highlighting its role in electrophysiological regulation [52]. Furthermore, NCAM-1 undergoes polysialylation, a post-translational modification required for proper sarcolemmal protein, and loss of this modification results in conduction defects [52]. Together, these findings suggest that NCAM signaling may facilitate early CM–fibroblast interactions before becoming specialized in conduction system cells to support synchronized ventricular activation.

Finally, the VEGF pathway displayed new signals between CMs and VEC, Art.EC in **Figure 8D**, consistent with spatial cell communication analysis of the postnatal murine heart by Wang et al., [51]. Its upregulation highlights its importance in angiogenesis and vascular remodeling, in line with

its roles in supporting coronary vessel formation and metabolic maturation during heart development.

5. Conclusion

In conclusion, in the current study, we successfully clustered and annotated all cell types in neonatal heart tissue using Seurat and explored biological pathways associated with different stages of CM development through differential gene set enrichment analysis. Neonatal cardiomyocytes progress from an immature proliferative state to metabolically and electrophysiologically mature cells through coordinated transcriptional programs and dynamic intercellular interactions. Early-stage CMs (D2D4) express cell-cycle and fetal myofibril genes (***Mki67***, ***Tnni1***, ***Ptprg***, ***Cox7a2***), while medium-stage CMs (D9D11) upregulate ion-handling, calcium-regulatory, and metabolic genes (***Trpc4***, ***Hcn2***, ***ATP2a3***) indicative of functional maturation. Furthermore, we identified several signaling pathways and their corresponding ligand–receptor pairs of stage-specific crosstalk with non-cardiomyocytes that may represent potential mechanisms driving cardiomyocyte maturation using CellChat. Main findings include macrophage-derived **ApoE–Trem2/Tyrobp** signaling promote early proliferation, angiogenesis, fibroblast-mediated **NCAM1–FGFR1** interactions contribute to early sarcomere assembly and structural organization, and **VEGF**-mediated endothelial signaling emerges later to facilitate vascular integration and maturation. Overall, these findings reveal the transcriptional and intercellular mechanisms orchestrating CM maturation and provide potential targets for regenerative therapies and functional cardiac tissue engineering.

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